

Kelson Qu

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Backend engineer with Java / Python / JavaScript, focused on Distributed systems, AgenticAI systems, and Cloud.

Education

Northeastern University – M.S. in Computer Science, GPA: 3.9/4.0 | San Jose, CA Sep 2023 – Aug 2026

- Coursework: Algorithms and Data Structure, Database Management, Computer Networks, Cloud Computing

Experience

Northeastern University, MIG Lab – Research Assistant | San Jose, CA Jan 2025 – Present

- HuskySQL: Built a natural-language Text-to-SQL system in **Python** on **BIRD** with a Divide-and-Conquer Chain-of-Thought **LLM** pipeline, classifier, and dual-pass workers, raising accuracy from 48% to 57% (+9pp).
- MRI Segmentation Pipeline: Engineered a **PyTorch** training and evaluation pipeline for 7 deep learning models on a 4,800-image dataset, deployed to a Boston clinical partner for initial screening of 200+ cases.

Evenness Inc. – Backend Engineer Intern | Cupertino, CA July 2025 – Dec 2025

- Compliance Scanner: Built **Python/FastAPI** WCAG 2.2 scanner combining **axe-core** rules with **LLM** fallback; **Redis** content-hash cache with per-rule TTL hit ~70%, cutting LLM cost ~60% on repeats.
- Auth & Access Control: Implemented **TypeScript/Express** multi-tenant auth with short-lived **JWT**, rotating refresh tokens, tenant-scoped **RBAC**, and **Zod** schemas; isolation enforced at token and query layers.
- Release & Observability: Hardened **GitHub Actions CI/CD** with **Docker** integration tests and staged rollout; instrumented **OpenTelemetry** tracing across 3 services to GCP Cloud Trace, cutting MTTR ~60%.
- Cloud Development: Migrated an internal API service from **AWS** to **GCP** under senior review; implemented **Cloud Run** deploy, **Firestore** (NoSQL) data layer, and zero-downtime cutover via weighted traffic splitting.

BMW Group – R&D Engineer Intern, New Tech R&D Center | Shanghai, China July 2022 – May 2023

- Benchmark Dashboard: Built a full-stack **React/Flask/InfluxDB** dashboard tracking time-series performance for 30+ hardware components, serving sub-200ms queries to support engineering review.
- Query Optimization: Optimized join-heavy analytics queries on **SAP HANA** with **materialized views** and predicate restructuring, cutting p95 latency 12s→5s.
- Search Service: Designed **RBAC**-protected search in **Django/PostgreSQL** with GIN-indexed full-text search over 300+ supplier records, replacing manual workflow with sub-2s lookups.

CDP Group – Software R&D Intern, Development Team | Shanghai, China July 2021 – Oct 2021

- Built and maintained attendance location and payroll upload features in **Java/SpringBoot/MySQL** on CDP's WorkLife HR app, replacing manual workflows with idempotent submissions and transactional CSV ingestion.

Projects

Multi-Agent LLM Asset Analysis Platform Java, Spring Boot, Spring AI, GCP, PostgreSQL, Redis, Terraform

- Architected a **cloud-native multi-agent** LLM investment platform on **GCP** with API and worker **microservices** for automated weekly asset briefings and real-time alerts across 20+ brokerages via SnapTrade **OAuth 2.0**; integrated Anthropic Claude via **Spring AI**.
- Evaluated **Apache Kafka** vs **GCP Pub/Sub** and adopted Pub/Sub push with **OpenID Connect (OIDC)** and **Dead-Letter Queue (DLQ)**; implemented **PostgreSQL** event sourcing for debate replay and **Redis** portfolio-hash cache, cutting LLM cost 40%.
- Parallelized 4 analyst agents with **Java 21 Virtual Threads**, reducing per-ticker time by 40%; secured **REST** and **Server-Sent Events (SSE)** APIs with **Spring Security** and **JWT**-based **RBAC**.
- Automated **Infrastructure as Code (IaC)** on **GCP** with **Terraform**, **Jib** distroless images, **Docker**, and **GitHub Actions**; instrumented **OpenTelemetry** traces to **Cloud Logging** and **Cloud Monitoring**.

Skills

Languages: Java, Python, TypeScript, JavaScript, SQL, Bash

Frameworks & Backend: SpringBoot, FastAPI, Express, Django, Flask, gRPC, REST, WebSocket, Microservices

Databases & Messaging: MySQL, PostgreSQL, SAP HANA, InfluxDB, Redis, Firestore, Kafka, RocketMQ

Cloud & DevOps: AWS, GCP, Docker, Linux, Git, GitHub Actions, CI/CD, OpenTelemetry, Grafana